In *Interstellar*, as the Ranger approaches Miller's planet carrying Cooper and his crew, we see Gargantua in the sky above, 10 degrees across (twenty times larger than the Moon as seen from Earth!) and surrounded by its bright accretion disk. See Figure 17.9. As startlingly impressive as this may be, Gargantua's angular size has actually been reduced greatly from what it would really be at the location of Miller's planet.

If Miller's planet is, indeed, close enough to Gargantua to experience extreme time slowing—as I chose for my interpretation of the movie—then it must be deep into the cylindrical region of Gargantua's warped space, as depicted in Figure 17.1. It seems likely, then, that if you look down the cylinder from Miller's planet you will see Gargantua, and if you look up the cylinder you will see the external universe; so Gargantua should encompass roughly half of the sky (180 degrees) around the planet and the universe the other half. Indeed, that is what Einstein's relativistic laws predict.

It also seems clear that, since Miller's planet is the closest anything can live stably, without falling into Gargantua, the entire accretion disk should be outside the orbit of Miller's planet. Therefore, as the crew approach the planet, they should see a giant disk above them and a giant black-hole shadow below. Again, that is what Einstein's laws predict.

If Chris had followed these dictates of Einstein's laws, it would have spoiled his movie. To see such fantastic sights so early in the movie would make the movie's climax, when Cooper falls into Gargantua, visually anticlimactic. So Chris consciously saved such sights for the end of the movie; and invoking artistic license, near Miller's planet he depicted Gargantua and its disk together, "just" twenty times bigger than the Moon looks from Earth.